

DAVIDE ITALO SERRAMAZZA

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Professional Summary

PhD in Explainable AI (XAI) for Time Series with 7+ first-author publications and a software library (tsCaptum) on PyPI. Proven ability to bridge academic research and production-ready code. Seeking research scientist or ML engineer roles requiring deep expertise in model interpretability, time series analysis, and PyTorch.

Education

Bachelor in Computer Science

Università di Pisa

September 2013 – September 2017

Pisa, Italy

Master in AI (School of Computer Science), Grade 110/110

Università di Pisa

September 2017 – March 2020

Pisa, Italy

PhD in XAI for Time Series

University College Dublin (UCD)

September 2022 – September 2026

Dublin, Ireland

Experience/Projects

University College Dublin

PhD on Time Series

September 2022 – September 2026 (thesis submission)

Dublin, Ireland

- Adapted XAI method as SHAP, Lime, Feature Ablation for Time Series
- Developed **tsCaptum**, a Python library bridging XAI library (Captum) with Time Series library (aeon, sktime). Computational time reduced by 1-2 orders of magnitude, performance not hurt.
pypi.org/project/tsCaptum/
- Published 4 papers as first author
- *Technical stack:* Python, PyTorch, Captum, aeon, sktime, scikit-learn, GitHub

Università di Pisa / Mediateca Toscana

Assegno di Ricerca (Research job)

May 2020 – May 2022

Tuscany, Italy

- Built an **API + database** to serve ML-generated video clips to a web platform
- Processed 100+ hours of documentary clips footage using OpenCV and FFmpeg
- *Technical stack:* Python, PyTorch, TensorFlow, GitHub, Node.js, MongoDB, REST APIs

Technical Skills

- **Programming Languages:** Python, Nodejs, Matlab, Java, JavaScript, C++, C, Bash script
- **AI Frameworks:** Pytorch, TensorFlow, sktime, aeon, sklearn, Captum
- **XAI:** SHAP, KernelSHAP, LIME, Feature Ablation, dCAM, frameworks (Captum, SHAP etc.)
- **Technologies/Frameworks:** Linux, GitHub, Docker, CUDA acceleration

Languages

- **Italian** (native)
- **English** (Professional working proficiency)